

Industry Daily Report — Pipeline Structure v3

The system as deployed and verified, 27 Aug 2026 · every number below is read from the live database, not estimated · companion to Methodology v2 and the Source Checklist

1 · The structure now — five workflows, one clock each

every hour :00	SOURCE INGEST	68 sources → 5-tier ladder → articles table
as mail arrives	NEWSLETTER INTAKE	mailbox → headline rules → same table
every 30 min	ARTICLE ANALYSIS	new rows → Gemini scores 0-100 (exits if none)
06:00 · 12:00 · 23:00	NEWS QUALITY	enrich dates → judge news/reference → coverage line
05:00	DAILY REPORT	window+ledger → 4 gates → render → Graph send → Teams card 05:20 · dashboard reads everything live

WORKFLOW	SCHEDULE (AEST)	JOB
BEXT — Source Ingest	hourly :00	Every active source down the retrieval ladder; every tier attempt logged
BEXT — Newsletter Intake	on mail arrival	Tier 0 — the lawful route past account walls (Reuters, The Australian, IEA)
BEXT — Article Analysis	every 30 min	Scores new articles; exits instantly when there is nothing new, so the cadence is free
BEXT — News Quality	06:00 / 12:00 / 23:00	The quality pass: dates, judgement, coverage. The 23:00 run closes the day
BEXT — Daily Report	05:00	Assembles the finished prior day, sends via Microsoft Graph

Plus: Daily News Card (05:20, Teams), Self-Heal (15 min), Graph Health (06:00), Contract Test (02:00), Meeting Intake, Teams Inbound, and the four content workflows. 15 total, all mapped live on the dashboard's architecture page.

2 · Hourly ingestion — how a source is actually fetched

One rule governs the ladder: the cheapest tier that demonstrably works, verified by evidence, visible when it fails. Every tier writes an attempt row every run — tried, skipped, succeeded, refused — so "was this even attempted?" is always answerable. That log is what caught every silent failure this week.

TIER	ROUTE	APPLIES TO	PROVEN ON
0	Email — newsletters into the intake mailbox, headline-rule extraction	Account-walled publishers	Reuters x5 briefings, The Australian, Conversation arriving daily
1	Scrapling — Chrome TLS fingerprint impersonation	Everything first; defeats WAF 403s	DCCEEW, NABERS, EcoGeneration, S&P: 403 → 200
1b	Sitemap — publisher's XML read as a feed, staggered so siblings never burst	Client-rendered listings with open sitemaps	S&P latest-news + crude-oil: 21 articles day one
2	Firecrawl — hosted JS rendering (metered: 04/10/16/22 only)	firecrawl: true sources	vic-premier 9 releases, aer-registers 21 decisions — after 428/494 straight failures
2	Chromium — local browser render	Ordinary JS pages (non-firecrawl)	standing fallback
3	Session login	Member portals (CEC built, credential pending)	—
4	Hermes reads the page	Markup no parser recognises	bounded last resort

Politeness is load-bearing: never two requests in flight to the same host, 1.2 s spacing — a flat pool earned 429s from our highest-volume source, and a 429 is indistinguishable from "no data" unless recorded apart (blocked , retried; never written off).

3 · The four causes — where each stands

CAUSE (25 AUG DIAGNOSIS)	STATUS	THE FIX, AND THE LIVE EVIDENCE
1. We never read the article — dated by fetch time; 232/318 undated; 1 in 3 filed under the wrong day	FIXED	Date read from the URL at ingest, then from the article page itself (article:published_time , JSON-LD) by News Quality; visited once (date_state). Live: 203 of 234 articles in the last 24 h carry the publisher's own date — 87%, from ~27%.
2. Hourly ingest drifts across midnight — evening stories fell between two	FIXED — structural	Window runs on the real publication date, two days back to now; the report_items ledger guarantees exactly-once. Verified: 0 duplicates, 0 losses since deployment. No per-source action exists or is

reports		needed.
3. Half of what we capture isn't an article — 429s mistaken for no-date; section pages as news	FIXED	Per-host politeness (429 → blocked → retried); Hermes news-vs-reference judge (10/10 on live titles, 33+ pages held); furniture rule (no date <i>and</i> score 0). Undecided ships — dropping a real story is the failure being engineered out.
4. Monitoring records but never alerts	PARTLY FIXED	Coverage line per quality pass (fetched · dated · undated · wrong-day-corrected · quiet sources); fetch_attempts retention 30 d; failed sends can no longer be recorded as sent. Still open: no Teams page when a source goes quiet — the count is measured, nobody is woken. Next build item.

4 · News Quality — the three-stage pass (06:00 / 12:00 / 23:00)

STAGE	WHAT IT DOES	GUARD RAILS
Enrich	Opens articles still undated, reads the publisher's own date, writes published_at + date_state	Per-host politeness · 4-min budget · a resolved date older than 14 days holds the article automatically (archive can never masquerade as news — 484 caught once)
Judge	Hermes separates news from standing pages (DWGM, ISP, "IT change & release management")	Batches of five (larger drifts out of format) · undecided ships · verdict stored, judged once
Monitor	Writes the coverage line to integration_health — the number to quote when asked "how many articles were affected"	Counts the Self-Heal blind spot: sources that ran clean and returned zero

The 23:00 pass is the reconciliation gate. The original design placed a catch-up step inside the 05:00 report; it was deliberately moved out — the send is the client deliverable, and a diagnostic that throws must not take it down (four consecutive runs were once lost that way). The 23:00 pass completes the day *before* it closes, so 05:00 reads finished data instead of racing it; the widened window + ledger make the gate unnecessary on the send path itself.

5 · The report — four gates, then send

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window: real publication date, day-2 → now          (drift-proof)
minus  everything already sent                      (ledger – exactly once)
gate 1 relevance ≥ 1      noise kept, wrong-subject dropped (score 0 stays in the data)
gate 2 not a reference page                (judge verdict)
gate 3 not furniture      (no date AND score 0 – never was an article)
gate 4 not stale-dated    (report_eligible hold)
→ render → Microsoft Graph send (token as urlencoded form; non-202 THROWS)
→ report row 'sent' only on a real 202 – a failed send can never look successful
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6 · Source Doctor — honest status

COMPONENT	STATUS	DETAIL
Four-fault playbook (diagnosis ladder)	DOCUMENTED	Methodology v2 Part 4 — what fires automatically, what to force, which tool per symptom
Quiet-source measurement	LIVE	Counted every quality pass in the coverage line
Judge + date remedies	LIVE	Running inside News Quality three times daily
Automated diagnosis (robots→sitemap proposal, YAML suggestion to Teams)	NOT BUILT	Today this procedure is run by hand — it found and fixed S&P. Automating it is the remaining Doctor work
Quiet-source Teams page	NOT BUILT	The measured count needs a webhook post + incidents row. Highest-value small item left

7 · Verification — how the system is checked, continuously

- Preflight (n8n/preflight.js) — 30+ regression assertions before any deploy; every rule cites a documented regression.
- Contract Test — nightly 02:00: every scheduled workflow has an unconditional heartbeat, Code nodes parse, inlined libs are stripped of exports.
- Attempt log — every source × tier × run; 30-day retention; the dashboard tier-strip reads it live.
- Coverage line — per quality pass; trends the wrong-day number toward zero.
- Ledger check — already-sent articles can never reappear (0 duplicates verified).
- Send guard — non-202 throws; both silent-send incidents this week were caught by it, not by a reader.
- Live watches — first runs of new machinery are monitored to completion, not assumed (Firecrawl's first window, the S&P sitemaps, the quality passes — each caught a real fault this way).

Open items, stated plainly: vic-premier rendered 9 releases on its first Firecrawl run then empty twice (hydration flakiness — under watch, retry logic if it persists) · first newsletter arrivals against the fixed keep-step are tonight — kept>0 is the proof · first coverage line lands at 23:00 tonight (enum cast fixed today) · quiet-source Teams page and automated Doctor diagnosis not yet built · IEA newsletter not yet arriving · a2ep.org.au/news from the brief not yet registered (our A2EP source watches the sibling aidc.org.au site).

BEXT Consultancy · automation stack · 27 Aug 2026 · architecture map: dashboard /architecture (15 workflows, 45 edges, generated from the deployed workflows — the map cannot drift from reality) · diagrams: docs/diagrams/ incl. BEXT-News-Quality